

AI-Based Onion Quality Assessment

A Comprehensive Implementation Strategy for DoCA Procurement (PS ID: 26031)

1. Why We Actually Need This

The Government of India (via NAFED and NCCF) procures millions of tonnes of onions annually to maintain a national buffer stock and control retail inflation. Currently, the assessment of this bulk produce is entirely manual. A nodal officer stands before a massive tractor-trolley of loose onions and makes a **subjective visual estimate** of the quality percentage.

This subjectivity creates three critical failures:

- **Disputes:** Farmers frequently accuse officials of unfairly downgrading their crop to pay them less, leading to protests at mandis.
- **Corruption:** Middlemen can collude with inspectors to classify substandard stock as Grade A, leading to massive financial losses for the government when the onions rot in storage godowns.
- **"Top-Dressing" Scams:** Farmers often hide rotten or undersized onions at the bottom of the crate or pile, tricking a superficial visual inspection.

An AI-driven approach eliminates human bias, ensures objective grading, and creates an immutable digital audit trail tied to GPS coordinates and timestamps.

2. Legal Framework & Government Specifications

The legal definition of agricultural quality in India is governed by the **Agricultural Produce (Grading and Marking) Act, 1937**, which established the **AGMARK** certification. For government buffer stock procurement, the Department of Consumer Affairs (DoCA) utilizes specific **FAQ (Fair Average Quality)** guidelines derived from AGMARK.

These laws mandate that quality is judged strictly on external, non-destructive visual parameters: equatorial diameter (size in mm), skin integrity (curing and scales), and the absolute absence of sprouting or fungal decay.

3. Grade A vs. URS (Under Relaxed Specifications)

Our computer vision model will be calibrated to segment and classify onions based on these strict government parameters:

Parameter	Grade A (Standard FAQ)	URS (Under Relaxed Specifications)
Equatorial Size	Strictly between 45 mm and 65 mm .	Relaxed to 35 mm to 70 mm .
Skin Integrity	Firm, well-cured, fully covered by dry, papery outer scales.	Partial skin peeling permitted; minor double-bulbs allowed.
Zero-Tolerance Defects	0% incidence of rot, black smut (fungus), sprouting, or mechanical damage.	Up to 10% allowance for minor sunburn. Rot and active sprouting remain strictly rejected.

Note: URS is temporarily invoked by the government during years of severe crop damage (droughts/floods) to prevent massive financial losses for farmers. The application must feature a toggle to switch between standard FAQ and URS threshold logic.

4. The Procurement Procedure (Ground Reality)

To build a practical solution, the software must integrate seamlessly into the harsh physical realities of an Agricultural Produce Market Committee (APMC) mandi.

1. **Arrival:** Farmers arrive with tractor-trolleys containing 20–30 quintals of loose onions.
2. **Unloading & Sorting:** The produce is poured onto large ground tarpaulins. Laborers manually sort (chhatayi) to remove glaringly obvious rotten or undersized bulbs.
3. **Inspection:** The nodal officer approaches the sorted pile to determine the final Grade A vs. Rejects percentage. **(This is where the app intervenes).**
4. **Weighing & Storage:** The approved stock is bagged in 50 kg jute sacks, tagged, and transported to storage godowns.

5. Our Implementation Strategy: The "Physical-Digital Handshake"

A simple photo-upload app will fail because it cannot evaluate the onions hidden beneath the top layer (occlusion). To win, we will implement workflow constraints that force comprehensive scanning.

Key Differentiator: Dynamic Video Sweeping

Instead of static photos, the app will mandate a **10-second video sweep** of the pile. The AI will extract random frames from the video. By tracking the onions as the inspector walks around the pile (or as the onions are actively poured), the system samples a vastly larger volume of the crop, effectively neutralizing the "top-dressing" scam.

Execution Steps:

- **AR-Guided Sampling:** The camera UI will display a 3x3 augmented reality grid over the pile, refusing to generate a report until the inspector captures footage from at least four distinct quadrants.
- **Reference Scaling:** To accurately measure diameter in millimeters, the app will require placing a standard object (e.g., an A4 clipboard or standard NAFED ID card) next to the pile to establish a dynamic pixel-to-millimeter ratio.
- **Edge Processing:** Since rural mandis have poor internet connectivity, the lightweight object detection model will run inference directly on the mobile device.
- **Immutable Audit Trail:** The final generated digital report will be cryptographically locked with the exact GPS coordinates of the mandi, timestamp, and device ID to prevent GPS spoofing or uploading pre-saved images.

6. Proposed Tech Stack

Our architecture focuses on offline-capable mobile edge computing paired with a robust centralized dashboard.

Frontend Application (Mobile)

- **Framework:** Kotlin (Native Android) for optimal camera performance and smooth UI on mid-range devices.
- **Local Database:** Room Database (SQLite) to store reports locally during offline network drops.
- **AR Integration:** ARCore for spatial grid overlays and reference scaling.

Computer Vision & AI

- **Model:** YOLOv8 (You Only Look Once) exported to TFLite format for rapid on-device inference.
- **Computer Vision:** OpenCV for Android (color space analysis to detect black smut and green sprouts).
- **Tracking:** DeepSORT algorithm to track individual onions across video frames and prevent double-counting.

Backend & Cloud

- **Framework:** Node.js with Express.
- **Database:** PostgreSQL with PostGIS for geospatial tracking of procurement reports.
- **Storage:** AWS S3 (or Firebase Storage) for archiving video evidence and final PDF reports.

Analytics Dashboard (Web)

- **Frontend:** React.js or Next.js.
- **Purpose:** A command center for DoCA officials in Delhi to monitor live procurement statistics, view dispute footage, and track anomaly flags across all national mandis.